

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Mechatronics Engineering

Course No.: MTE 3102

Course Title: Control Systems Sessional

Experiment No.: 02

Experiment Name: Design a PID, PI, PD controller for various systems by using trial and error method and analysis their performance

<u>Remarks</u>

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Date of Experiment: 12/1/2026

Date of Submission:

Experiment No.: 02

Experiment Name: Design a PID, PI, PD controller for various systems by using trial and error method and analysis their performance

Objectives:

- Design a mechanical system and control it using PID, PI, PD
- To design PID, PI, and PD controllers
- Simulate their time-domain responses using MATLAB, and to investigate their transient performance.
- To obtain the step response of a transfer function in control system and a mechanical system,
- To analyze their performance.

Required Equipment's:

- Computer with MATLAB software
- Necessary module for system modeling

Theory:

Any modern control systems must operate accurately smoothly, and efficiently. Likely we are in the era of modern technologies where we all are busy in building something and so everybody has high expectations on our buildings. And more likely we love to control and automate them. So we have control system that has two types open loop control system and closed loop control system. Nearly everything with us are open loop control system as we does control them with our hands or put a reference value and the controller reach that value manipulating the system. But mostly we are lazy to work for and so we does it automatically like electric vehicles(AVR), robotics, conveyors, and every automatic things. And this automatic things are made in closed loop control system which is exactly same as open loop control system but with a extra module feedback control. [1]

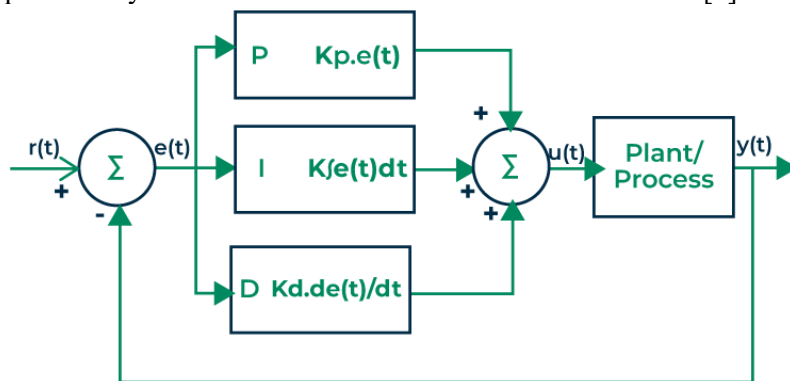


Fig 2.1: PID Controller [4]

A controller compares the desired references with the actual output of the system and generates a correction signal. The most widely used controller in Engineering is PID, PI and PD controllers. Each of this provides different performance characteristics in terms of stability, speed of response, and steady-state accuracy. [2] In PID P means proportional, I means integration and D means derivatives. Every value of PID is really important for a system and P decides system response or timing of a system if P increases system response faster, risetime decreases, overshoot and oscillation increases. For increasing the value of I error of steady-state reduces, output aligns with the reference value more, but oscillation increases, settling time also increases so does system becomes unstable. And if we increase the value of D overshoot decreases, oscillation decreases also the system becomes more smooth but on the other hand the system becomes noise sensitive, responses become slow. In short K_p for speed, K_i for accuracy and K_d for stability. [1] PI and PD are two sub types of PID where PI means the value of D is zero and for PD value of I is zero.

PID Controller Equation:

$$u(t) = K_p * e(t) + K_i \int e(t)dt + K_d * de(t)/dt$$

In Laplace Domain,

$$G(s) = K_p + K_i/s + K_d s$$

System Model:

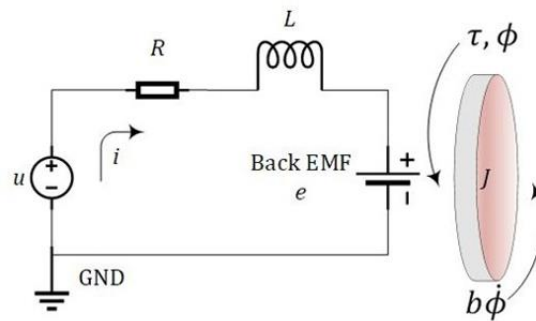


Fig 2.2: System Representation of DC motor [5]

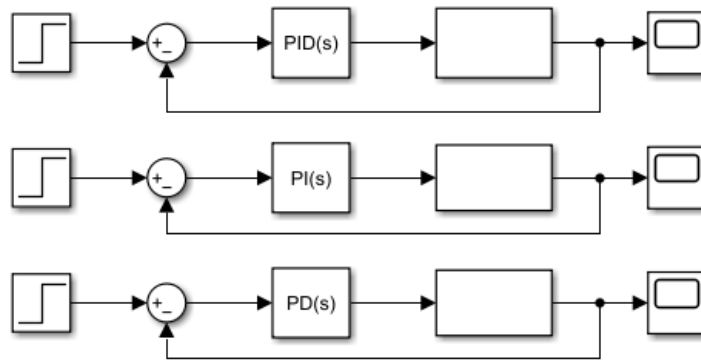


Fig 2.3: Closed loop system using PID, PI & PD in MATLAB Simulink

Transfer Function:

Simplified transfer function of a Mechanical system:

$$G(s) = \frac{k}{(Js + c)(Ls + r) + k^2}$$

Where:

- J = moment of inertia
- c = friction coefficient
- L = inductance
- r = resistance
- k = motor constant

Using the given values:

$$J = 0.01, c = 0.1, L = 0.5, r = 1, k = 0.01$$

The plant transfer function becomes:

$$G(s) = \frac{0.01}{0.005s^2 + 0.06s + 0.1001}$$

This equation represents how motor speed responds to applied voltage.

MATLAB Code:

```
clc
clear
% Motor parameters
j=0.01;
c=0.1;
k=0.01;
r=1;
l=0.5;
s=tf('s');
```

```

TF = k/((j*s + c)*(1*s + r) + k^2);
% Controller gains (use your tuned PID)
Kp = 6999;%2.0;
Ki = 999999;%16.99;
Kd = 699;%0.003;
PID = pid(Kp, Ki, Kd);
% Controller for PI
Kp = 120.8;
Ki = 120.99;
PI = pid(Kp, Ki, 0);
% Controller for PD
Kp = 100.0;
Kd = 18000.99;
PD = pid(Kp, 0, Kd);
% Closed loop systems
CL_PID = feedback(TF*PID,1);
CL_PI = feedback(TF*PI,1);
CL_PD = feedback(TF*PD,1);
% Plot all responses
figure(1)
hold on
grid on
step(TF,'k')
step(CL_PI,'g')
step(CL_PD,'r')
step(CL_PID,'b')
title('Open Loop vs PI vs PD vs PID Control')
legend('Open Loop','PI Control','PD Control','PID Control')
hold off
info = stepinfo(CL_PID)
info = stepinfo(CL_PI)
info = stepinfo(CL_PD)

```

Result:

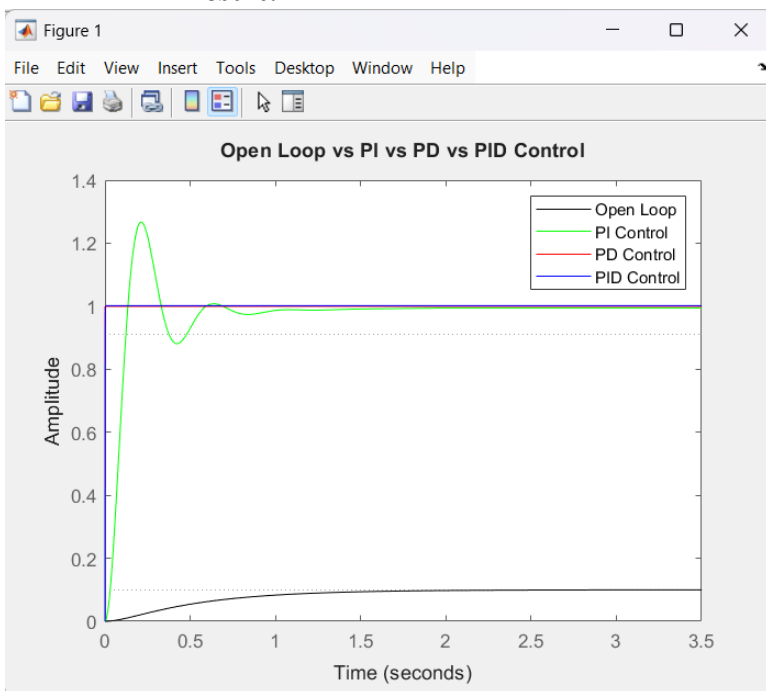


Fig 2.4: System response of PID, PI, PD & open loop control system together from MATLAB script

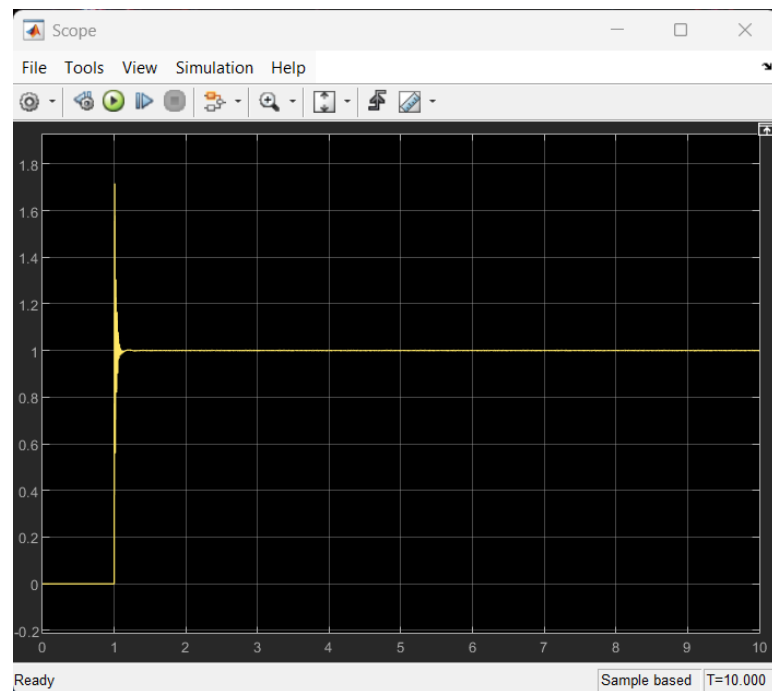


Fig 2.5: System response in PID controller from MATLAB Simulink

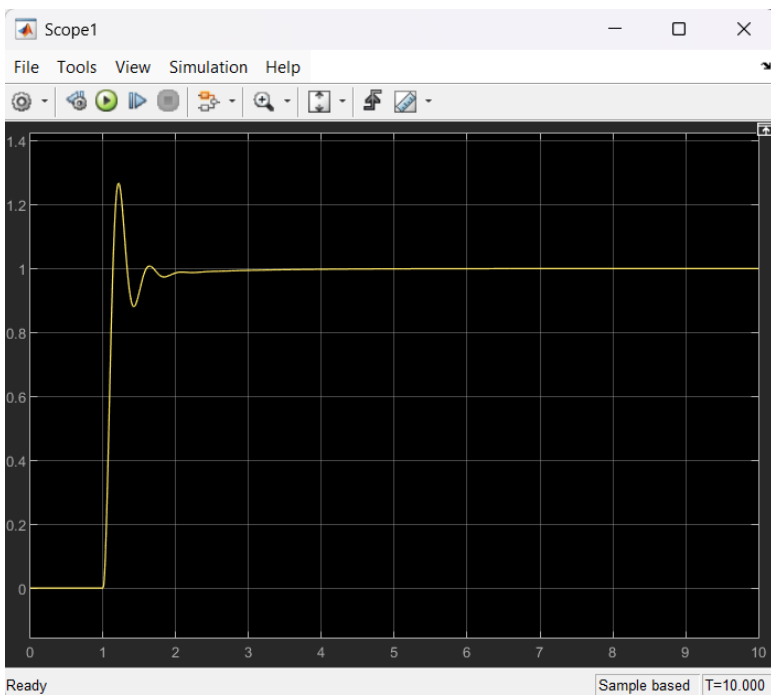


Fig 2.6: System response in PI controller from MATLAB Simulink

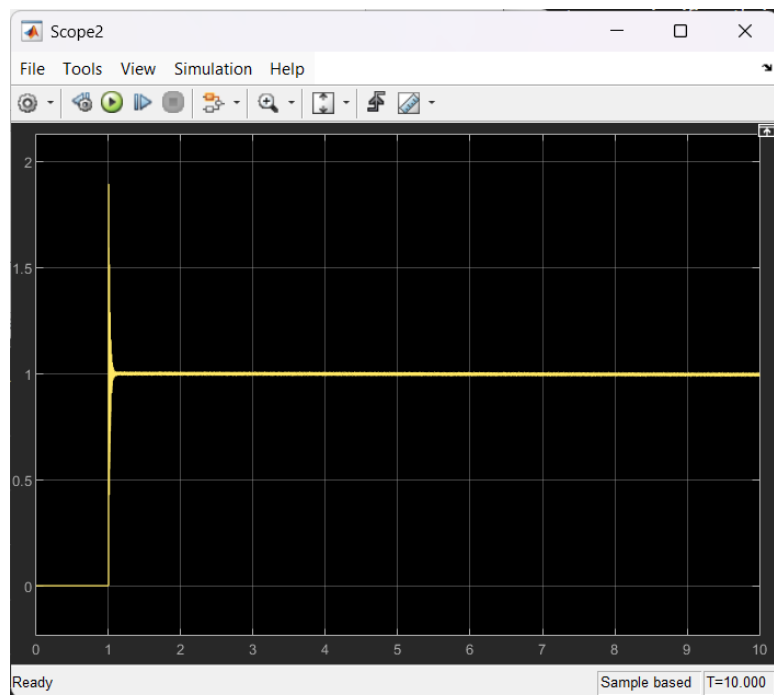


Fig 2.7: System response in PD controller using MATLAB Simulink

Performance Analysis:

Systems	Rise Time(s)	Peak Time (s)	%Overshoot	Settling Time(s)
PID	0.0016	0.0052	0.1725%	0.0028
PI	0.0911	0.2086	26.6970%	0.9271
PD	4.4718e-05	4.9998e-04	9.9733%	N/A

Discussion:

The open-loop motor shows slow and inaccurate response. The PI controller eliminates steady-state error but increases rise time. The PD controller improves speed but still has steady-state error. The PID controller provides the best response: fast rise time, minimal overshoot, and zero steady-state error. In every case if we see PI it has only two parameters proportional and integration for which it's system response is well enough as well as rise time means it's accurate but high overshoot which says the system is not stable. If we look at PD it rise time is really low as well as peak time with no settling time which describes the PD controller is efficient and stable but it not that accurate. But among them we can say PID controller is best because it includes all means the system is stable reliable accurate with low rise time and settling time and best among this are it overshoot is 0.1725% which is really low.

Conclusion:

The simulation results clearly showed that each controller has distinct characteristics. The PI controller eliminated steady-state error but introduced large overshoot and slower settling, making it less suitable for high-performance applications. The PD controller provided very fast response and improved stability but failed to remove steady-state error. The PID controller, which combines proportional, integral, and derivative actions, delivered the best overall performance by achieving fast rise time, very low overshoot, zero steady-state error, and stable settling behaviour. Both MATLAB and Simulink results verified that PID control is the most effective strategy electric vehicles, robotics, and industrial automation, where accuracy, stability, and fast response are all critical.

References:

1. Nise, N. S., *Control Systems Engineering*, 7th Edition, Wiley.
2. MATLAB Documentation, *Control System Toolbox*, MathWorks.
3. Class notes and lab manual
4. [geeksforgeeks.com](https://www.geeksforgeeks.com)
5. collimator.com